TECHNICAL TRAINING CATALOGUE 2026

OUR COMMITMENT:

TO ASSIST YOU IN DEVELOPING THE COMPETENCE AND PERFORMANCE OF YOUR EMPLOYEES AND YOUR SERVICE.

Version of October 27, 2025)

EDITO

Dear Subsidiary,

The Group Academy is pleased to present its catalogue of services for 2026 and to inform you of our commitments for all subsidiaries:

- ✓ **To support you in developing the skills and performance of your employees and your service,**
- ✓ **To support you in resolving technical problems on customer vehicles,**
- ✓ **To standardize and normalize the skill levels of the group's technicians.**

The diversity of brands deployed by CFAO is a real asset for the development of our skills development offers to subsidiaries throughout the company's territories. This diversity challenges us to ensure a high level of service quality within the after-sales service, the parts service, the sales department and the financial and legal service.

To improve the quality of your employees' services, we recommend deploying the following technical skills acquisition cycle:

- ✓ **Measure** the current skills of technicians and teams,
- ✓ **Exploit** the results to guide the skills development strategy,
- ✓ **Define** individual and collective skills development plans,
- ✓ **Acquire** new skills through face-to-face or remote trainings,
- ✓ **Certify** the new skills acquired,
- ✓ **Apply** new skills in a professional environment,
- ✓ **Perform** in a professional environment.

We also offer to **assist** you in **resolving technical problems** that you encounter in your workshops, with **technical support integrated** into the Academy.

The development of new skills is carried out by certified experts each in their areas of expertise, using innovative training methods.

Thank you for your commitment to developing the skills of your employees. We look forward to hearing from you.

Laurent GONIN

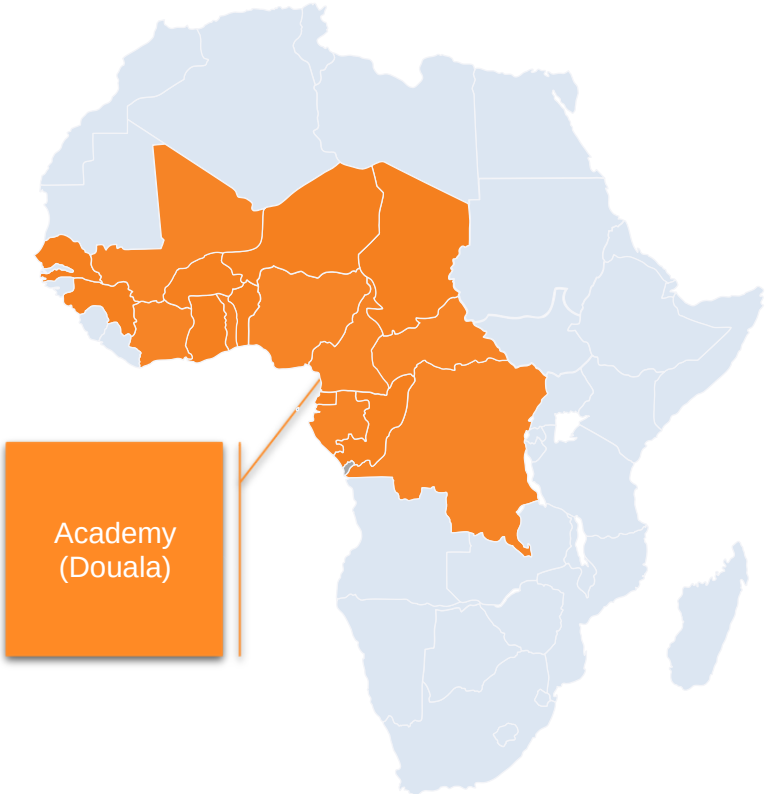
TEAM EQUIPMENT

The Douala Academy for EQUIPMENT is dedicated to the RENAULT TRUCKS, JCB, LOVOL, TOYOTA FORKLIFT, BT, FUSO, HINO, SINOTRUK and KING LONG brands and is equipped with:

- A reception and a restaurant,
- Two training rooms (board, video projector, others),
- A 200 m² workshop,
- Didactic equipment for engines, transmissions and others,
- A RENAULT TRUCKS K Range truck

Our training courses are delivered at the Douala Academy and at all our subsidiaries. We also provide training at our customers' sites. Training sessions are made up of a maximum of twelve trainees.

Training Contact service:
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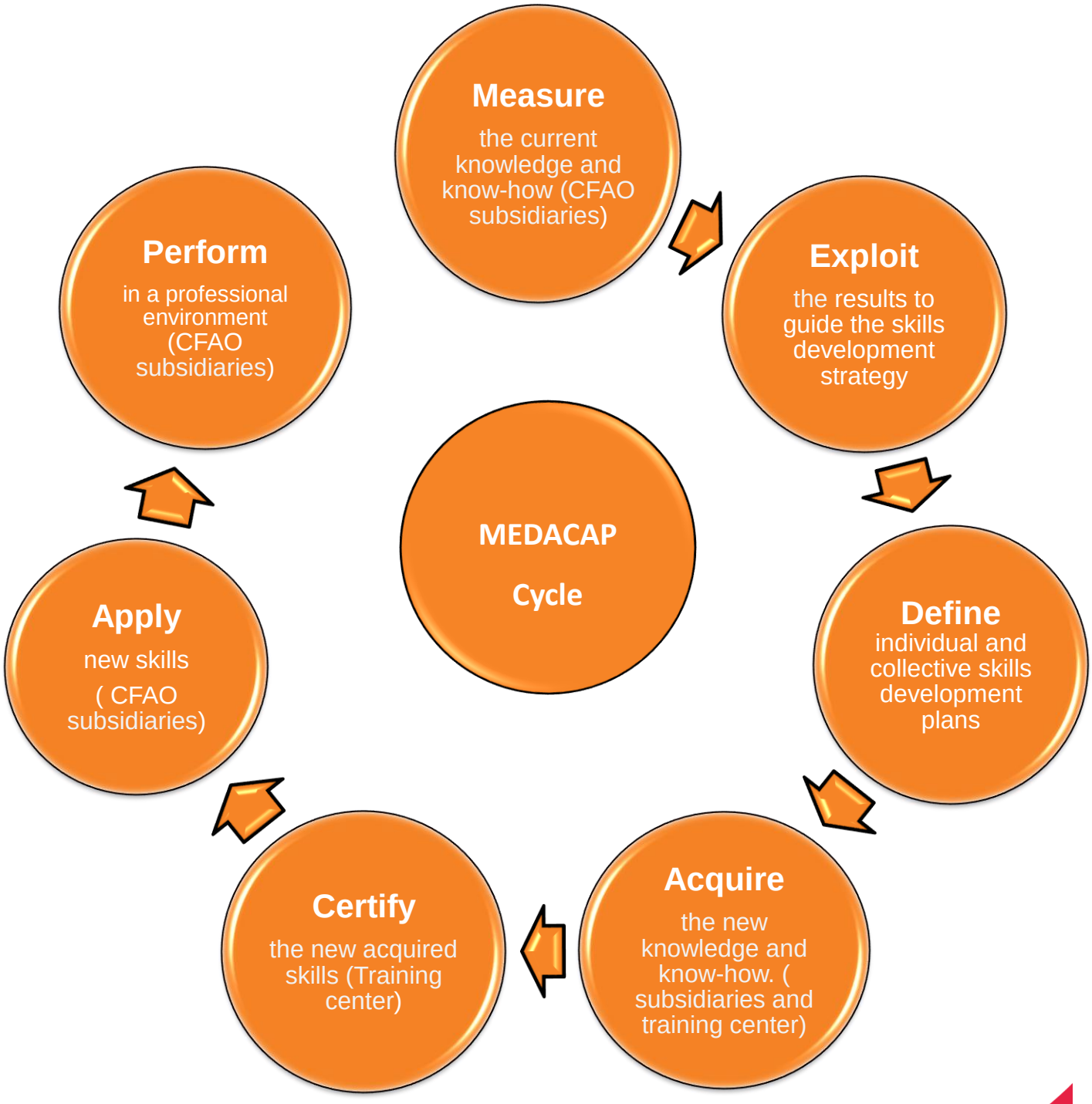


Thierry KOUGHA
Technical Trainer Expert (TOYOTA FORKLIFT, JCB, LOVOL)

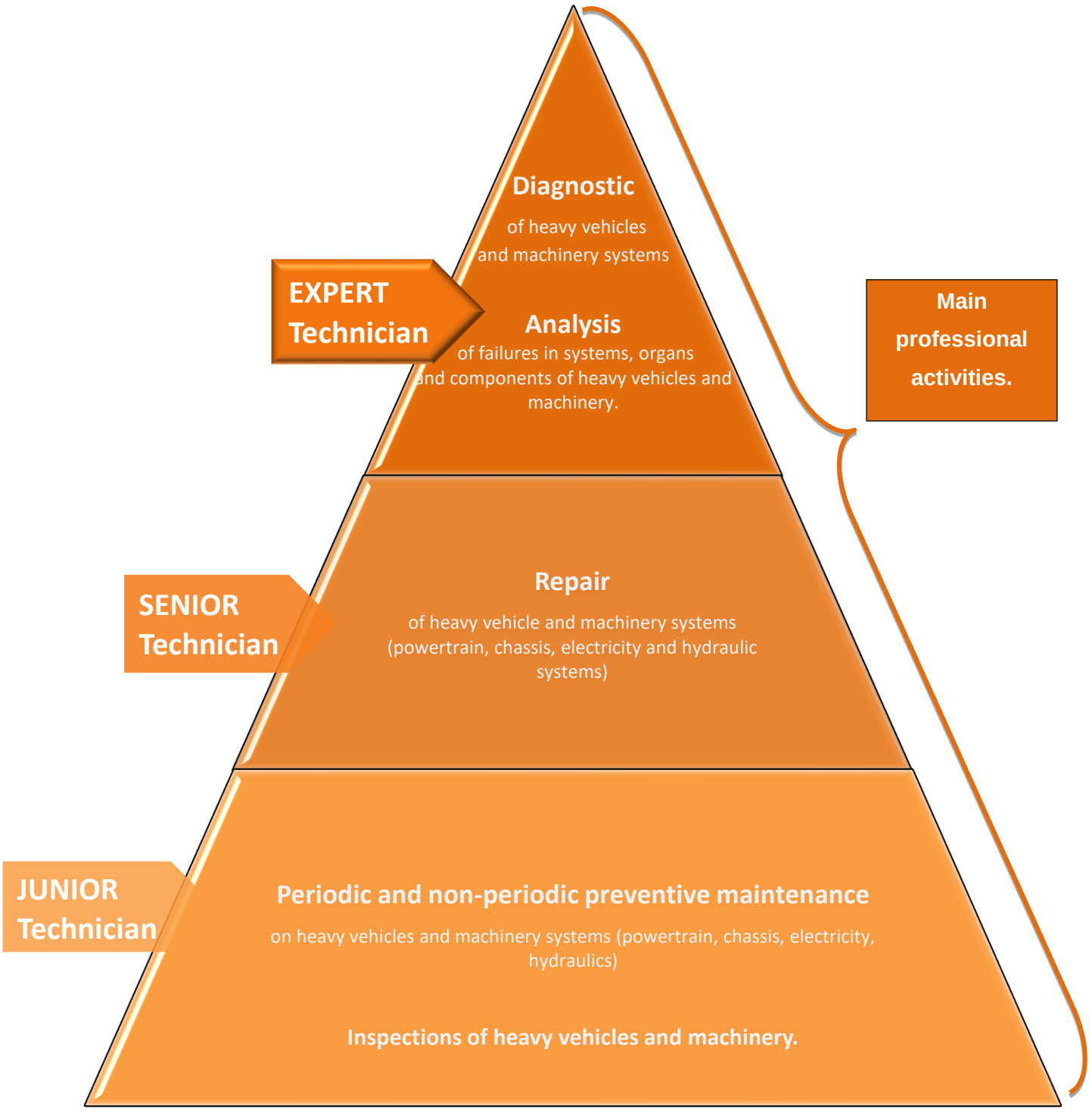


Roland NENDA
Operator Trainer All- brands expert (Trucks and Heavy Equipment)

DEVELOPING TECHNICIANS' PROFESSIONAL EXPERTISE



SKILL LEVELS OF TECHNICIANS



TRAINING BY TECHNICIAN LEVEL

	Theoretical training (Knowledge - > E-Learning)		Practical training (Know-how -> Face-to-face)	
	Thématics	Duration (days)	Thématics	Duration (days)
EXPERT Technicien	❖ The 7 stages of diagnostics applied to different vehicle systems (powertrain, chassis, electricity, hydraulics), ❖ Functionalities of the Diag tool.	10	❖ Electrical/Electronic Diagnostic of electric forklift trucks, ❖ Electrical/Electronic Diagnostic of thermal forklift trucks.	5 5
SENIOR Technicien	❖ Technological knowledge of different vehicle systems (powertrain, chassis, electricity, hydraulics). ❖ Procedure for removal and refitting in accordance with the manufacturer's recommendations., ❖ Procedure for Disassembly and reassembly in accordance with manufacturer's recommendations. ❖ Procedure for checking components according to the manufacturer's recommendations.	30	❖ Repair of automatic and hydrostatic transmissions on thermal forklift trucks, ❖ Repair of traction units on electric forklift trucks. ❖ Repair of gasoline and gas hybrid engines, ❖ Repair of gasoline and gas hybrid engines (4Y-ECS with electronic management), ❖ Repair of Diesel 1ZS Engine, ❖ Repair of Diesel 1DZ Engine II-III, ❖ Repair of Diesel 3Z Engine, ❖ Hydraulics Repair of electric and thermal forklift trucks	5 5 5 5 5 5 5 5
JUNIOR Technicien	❖ Brand history (forklift ranges, characteristics, etc.), ❖ Description of the structure of the forklift and role of the different forklift truck systems (powertrain, chassis, electricity, hydraulics). ❖ Role of fluids and lubricants. ❖ Principles of use of metrology tools, ❖ Periodic and non-periodic preventive maintenance of forklift trucks, ❖ Fundamental principles of the engine, transmission, electricity and hydraulics.	15	❖ Preventive maintenance of thermal forklift trucks 8FG(D)10-35, ❖ Preventive maintenance of thermal forklift trucks 8FG(D)40-80, ❖ Preventive maintenance of thermal forklift trucks 4(5) FD100-240, ❖ Preventive maintenance of electric forklift trucks 8FB10-35.	5 5 5 5

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Page	Code	Training designation	Date	Number of days	Place
8 to 12	JUNIOR TECHNICIAN (PREVENTIVE MAINTENANCE)				
9	TOY01	Preventive maintenance of thermal trucks 8FG(D)10-35,	To be defined	5 days	To be defined
10	TOY02	Preventive maintenance of 8FG(D)40-80 thermal trucks,	To be defined	5 days	To be defined
11	TOY02	Preventive maintenance of 4(5) FD100-240 thermal trucks,	To be defined	5 days	To be defined
12	TOY04	Preventive maintenance of 8FB10-35 electric trucks.	To be defined	5 days	To be defined
13 to 21	SENIOR TECHNICIAN (REPAIR)				
14	TOY05	Repair of automatic and hydrostatic transmissions for thermal trucks,	To be defined	5 days	To be defined
15	TOY06	Repair of the traction unit for electric forklifts. To be defined. To be defined.	To be defined	5 days	To be defined
16	TOY07	Hybrid Gasoline and Gas Engine Repair,	To be defined	5 days	To be defined
17	TOY08	Hybrid gasoline and gas engine repair (4y-ecs electronically managed)	To be defined	5 days	To be defined
18	TOY09	1ZS Diesel Engine Repair	To be defined	5 days	To be defined
19	TOY10	1DZ II-III Diesel Engine Repair	To be defined	5 days	To be defined
20	TOY11	3Z Diesel Engine Repair	To be defined	5 days	To be defined
21	TOY12	Hydraulic repair of electric and thermal carts.	To be defined	5 days	To be defined
22 to 24	EXPERT TECHNICIAN (DIAGNOSTICS)				
23	TOY13	Electrical/electronic diagnostics of electric trucks,	To be defined	5 days	To be defined
24	TOY14	Electrical/electronic diagnosis of thermal trucks.	To be defined	5 days	To be defined

JUNIOR TECHNICIAN

PREVENTIVE MAINTENANCE



COUNTERBALANCE TRUCKS 4FD100-240



COUNTERBALANCE TRUCKS 8FG(D)40-80



COUNTERBALANCE TRUCKS



COUNTERBALANCE TRUCKS 8FB1.5

TOY01	PREVENTIVE MAINTENANCE OF THERMAL TROLLEYS Series: 8FG(D)10-35

Level, mode & duration	JUNIOR	Face-to-face	05 days
Trainer	**Thierry KOUGHA**		
Goals	• **Maintain** the systems (powertrain, chassis, electricity, hydraulics) in operation according to the manufacturer's recommendations, • **Inspect** the systems (powertrain, chassis, electricity and hydraulics) of the machine according to the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical Part:** • 8FG(D)10-35 Forklift Ranges and their Features, • Description of the truck systems (Powertrain, Chassis, Electricity, Hydraulics, Masts, etc.) and their components, • Description of the truck's safety and stability systems (OPS and SAS), • The different types of maintenance applied to 8FG(D)10-35 forklifts and maintenance periods in accordance with the manufacturer, • Periodic and Non-periodic Preventive Maintenance of systems and components in accordance with the manufacturer. ► **Practical Part :** • Replace consumables (Engine, Transmission, Hydraulics, etc.), • Top up oils and fluids (Engine, gearbox, axles, hydraulic systems and others), • Check the conformity of the components (dimensional, flatness, voltage, intensity, sealing, pressure, etc.) according to the manufacturer's recommendations, • Remove and replace components (filters, belts and accessories, cylinders, hydraulic hoses, etc.) according to the manufacturer's recommendations, • Visually inspect the systems of the 8FG(D)10-35 Thermal Carts, • Start and reset the maintenance counter after work in accordance with the manufacturer, • Prepare the trucks before delivery (PDI: Pre-delivery Inspection) in accordance with the manufacturer. ► **Evaluation Part:** • Initial Test (Assessment of Prerequisites). • Final Test (Assessment of Acquired skills).		
Target audience	Operational technicians in preventive maintenance		
Prerequisites	E-learning and/or face-to-face training in basic mechanics, electricity and hydraulics		
Language	French		
Equipment	Fully-equipped training room and workshop, adapted forklift truck(s) available, SST		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation		

TOY02	PREVENTIVE MAINTENANCE OF THERMAL FORKLIFTS TRUCKS 8FG(D)40-80

Level, mode & duration	JUNIOR	Face-to-face	05 days
Trainer	**Thierry KOUGHA**		
Objectives	• **Maintain** the systems (powertrain, chassis, electricity and hydraulics) in operation in accordance with the manufacturer's recommendations, • **Inspect** the machine's systems (powertrain, chassis, electricity and hydraulics) in accordance with the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical part:** • Front forklift trucks 8FG(D) 40-80 Ranges and their characteristics, • Description of the forklift systems (Powertrain, Chassis, electricity, hydraulics, Masts, ...) and their components, • Description of forklift safety and stability systems (OPS and SAS), • The different types of maintenance applied to 8FG(D)40-80 forklift trucks and maintenance periods in accordance with the manufacturer's recommendations, • Periodic and non-periodic preventive maintenance of systems and components in compliance with the manufacturer's recommendations. ► **Practical part:** • Replace consumables (Engine, Transmission, Hydraulics, ...), • Replenish oils and fluids (Engine, BV, axles, hydraulic systems and others), • Check component conformity (dimensions, flatness, voltage, current, tightness, pressure, etc.) in accordance with the manufacturer's recommendations, • Remove and refit components (filters, belts and accessories, jacks, hydraulic hoses, etc.) in accordance with the manufacturer's recommendations, • Visually inspect the 8FG(D)40-80 thermal forklift truck systems, • Start up and reset the maintenance counter after work in accordance with the manufacturer's recommendations, • Prepare forklifts prior to delivery (PDI: Pre-delivery Inspection) in accordance with the manufacturer's recommendations. ► **Evaluation part:** • Initial Test (Assessment of Prerequisites). • Final Test (Assessment of Acquired skills).		
Target audience	Operational technicians in preventive maintenance		
Prerequisites	E-learning and/or classroom training in Mechanics, Electricity and Basic Hydraulics		
Language	French		
Equipment	Fully-equipped training room and workshop, adapted forklift truck(s) available, SST		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation		

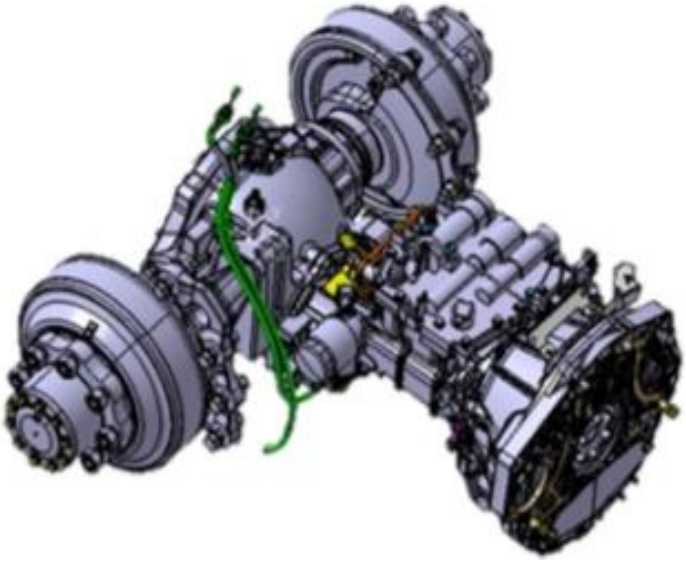
TOY02	PREVENTIVE MAINTENANCE OF THERMAL FORKLIFTS TRUCKS 4(5) FD100-240

Level, mode & duration	JUNIOR	Face-to-face	05 days
Trainer	**Thierry KOUGHA**		
Objectives	• **Maintain** the systems (powertrain, chassis, electricity and hydraulics) in operation in accordance with the manufacturer's recommendations, • **Inspect** the machine's systems (powertrain, chassis, electricity and hydraulics) in accordance with the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical part:** • Front forklift trucks 4(5) FD100-240 Ranges and their characteristics, • Description of the forklift systems (Powertrain, Chassis, electricity, hydraulics, Masts, ...) and their components, • Description of forklift safety and stability systems (OPS and SAS), • The different types of maintenance applied to 4(5) FD100-240 forklift trucks and maintenance periods in accordance with the manufacturer's recommendations, • Periodic and non-periodic preventive maintenance of systems and components in compliance with the manufacturer's recommendations. ► **Practical part:** • Replace consumables (Engine, Transmission, Hydraulics, ...), • Replenish oils and fluids (Engine, BV, axles, hydraulic systems and others), • Check component conformity (dimensions, flatness, voltage, current, tightness, pressure, etc.) in accordance with the manufacturer's recommendations, • Remove and refit components (filters, belts and accessories, jacks, hydraulic hoses, etc.) in accordance with the manufacturer's recommendations, • Visually inspect the 4(5) FD100-240 thermal forklift truck systems, • Start up and reset the maintenance counter after work in accordance with the manufacturer's recommendations, • Prepare forklifts prior to delivery (PDI: Pre-delivery Inspection) in accordance with the manufacturer's recommendations. ► **Evaluation part:** • Initial Test (Assessment of Prerequisites). • Final Test (Assessment of Acquired skills).		
Target audience	Operational technicians in preventive maintenance		
Prerequisites	E-learning and/or classroom training in Mechanics, Electricity and Basic Hydraulics		
Language	French		
Equipment	Fully-equipped training room and workshop, adapted forklift truck(s) available, SST		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation		

TOY04	PREVENTIVE MAINTENANCE OF ELECTRIC FORKLIFT TRUCKS 8FB10-35

Level, mode & duration	JUNIOR	Face-to-face	05 days
Trainer	**Thierry KOUGHA**		
Objectives	• **Maintain** the systems (AC motor, chassis, electricity, hydraulics) in operation in accordance with the manufacturer's recommendations, • **Inspect** the machine's systems (AC engine, chassis, electricity and hydraulics) in accordance with the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical part:** • Electric Forklift Truck 8FB10-35 ranges and their characteristics, • Description of the truck systems (AC engine, Chassis, electricity, hydraulics, Masts, battery charging equipment, etc.) and their components, • Description of the forklift's safety and stability systems (OPS and SAS), • The different types of maintenance applied to the 8FB10-35 forklift trucks and maintenance periods in accordance with the manufacturer's recommendations, • Periodic and non-periodic preventive maintenance of systems and components in compliance with the manufacturer's recommendations, ► **Practical part:** • Replace consumables (Batteries, Transmission, Hydraulics, ...), • Replenish oils, fluids and greases (Battery, Axles, Hub, Hydraulic systems, Masts and others), • Check the conformity of components (dimensions, flatness, voltage, current, tightness, pressure, etc.) in accordance with the manufacturer's recommendations, • Remove and refit components (filters, battery packs, jacks, hydraulic hoses, etc.) in accordance with the manufacturer's recommendations, • Visually inspect the systems on electric forklift trucks in the 8FB10-35 range, • Start up and reset the maintenance counter after work in accordance with the manufacturer's recommendations, • Prepare the trucks before delivery (PDI: Pre-delivery Inspection) in accordance with the manufacturer's recommendations. ► **Evaluation part:** • Initial test (Assessment of prerequisites). • Final Test (Assessment of Acquired skills).		
Target audience	Operational technicians in preventive maintenance		
Prerequisites	E-learning and/or On-site training in Mechanics, Electricity and Basic Hydraulics		
Language	English - French		
Equipment	Fully-equipped training room and workshop, Adapted forklift trucks(s) available, SST		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation		

SENIOR TECHNICIAN
REPAIR





TOY05	REPAIR OF AUTOMATIC AND HYDROSTATIC TRANSMISSIONS ON THERMAL FORKLIFT TRUCKS

Level, mode & duration	SENIOR	Face-to-face	05 days
Trainer	**Thierry KOUGHA**		
Objectives	• **Remove** and **refit** transmission components in accordance with the manufacturer's recommendations, • **Disassemble** and **reassemble** the gearbox, electrohydraulic unit and hydraulic motors in accordance with the manufacturer's recommendations, • **Check** component conformity in accordance with the manufacturer's recommendations, • **Test** and **adjust** the gearbox, electrohydraulic unit and hydraulic motors.		
Description Theory (40%) Practical (60%)	► **Theoretical part:** • Reminder of specific safety measures in the repair workshop, • Description and function of automatic and hydrostatic transmissions, • Architecture of the gearbox, electrohydraulic gear control unit and hydraulic motors, • Procedure for removal and refit of the gearbox, electrohydraulic unit and hydraulic motors in accordance with the manufacturer's recommendations, • Procedure for disassembly and reassembly of gearbox components, hydraulic unit and hydraulic motors in accordance with the manufacturer's recommendations, • Manufacturer's check procedure for engine components. ► **Practical part:** • Remove the torque converter, gearbox, hydraulic block and hydraulic drive motors in accordance with the manufacturer's recommendations, • Disassemble the gearbox, hydraulic block and hydraulic motors according to the manufacturer's recommendations, • Check that the components (solenoid valves, seals, clutch blocks, hydraulic connections, etc.) comply with the manufacturer's recommendations, • Replace components that do not comply with the manufacturer's recommendations, • Reassemble components in accordance with the manufacturer's recommendations, • Refit the gearbox, hydraulic unit and hydraulic engine, • Test and adjust the gearbox, electrohydraulic unit and hydraulic engine. ► **Evaluation part:** • Initial Test (Assessment of prerequisites). • Final Test (Assessment of Acquired skills).		
Target audience	Technician carrying out repairs		
Prerequisites	Have acquired the skills of a JUNIOR Technician (Preventive Maintenance)		
Language	French		
Equipment	Fully equipped training room and workshop, Adapted forklift truck(s) available, Bench-mounted components and special tools (SST)		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation.		

TOY06	REPAIR OF THE TRACTION UNIT FOR ELECTRIC FORKLIFTS.

Level, mode & duration	SENIOR	Face-to-face	05 days
Trainer	**Thierry KOUGHA**		
Objectives	• **Remove** and **refit** transmission components in accordance with the manufacturer's recommendations, • **Disassemble** and **reassemble** the gearbox, electrohydraulic unit and hydraulic motors in accordance with the manufacturer's recommendations, • **Check** component conformity in accordance with the manufacturer's recommendations, • **Test** and **adjust** the gearbox, electrohydraulic unit and hydraulic motors.		
Description Theory (40%) Practical (60%)	► **Theoretical part:** • Reminder of specific safety measures in the repair workshop, • Description and function of automatic and hydrostatic transmissions, • Architecture of the gearbox, electrohydraulic gear control unit and hydraulic motors, • Procedure for removal and refit of the gearbox, electrohydraulic unit and hydraulic motors in accordance with the manufacturer's recommendations, • Procedure for disassembly and reassembly of gearbox components, hydraulic unit and hydraulic motors in accordance with the manufacturer's recommendations, • Manufacturer's check procedure for engine components. ► **Practical part:** • Remove the torque converter, gearbox, hydraulic block and hydraulic drive motors in accordance with the manufacturer's recommendations, • Disassemble the gearbox, hydraulic block and hydraulic motors according to the manufacturer's recommendations, • Check that the components (solenoid valves, seals, clutch blocks, hydraulic connections, etc.) comply with the manufacturer's recommendations, • Replace components that do not comply with the manufacturer's recommendations, • Reassemble components in accordance with the manufacturer's recommendations, • Refit the gearbox, hydraulic unit and hydraulic engine, • Test and adjust the gearbox, electrohydraulic unit and hydraulic engine. ► **Evaluation part:** • Initial Test (Assessment of prerequisites). • Final Test (Assessment of Acquired skills).		
Target audience	Technician carrying out repairs		
Prerequisites	Have acquired the skills of a JUNIOR Technician (Preventive Maintenance)		
Language	French		
Equipment	Fully equipped training room and workshop, Adapted forklift truck(s) available, Bench-mounted components and special tools (SST)		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation.		

TOY07	REPAIR OF GASOLINE AND GAS HYBRID ENGINES

Level, mode & duration	SENIOR	Face-to-face	05 days
Trainer	**Thierry KOUGHA**		
Objectives	• **Remove** and **refit** the 1FS Gasoline and Gas Hybrid engine in accordance with the manufacturer's recommendations, • **Disassemble** and **reassemble** the 1FS gasoline and gas hybrid engine in accordance with the manufacturer's recommendations, • **Check** component conformity in accordance with the manufacturer's recommendations, • **Test** and **adjust** the 1FS gasoline and gas Hybrid engine in accordance with the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical part:** • Reminder of specific safety measures in the repair workshop, • Description and operation of the 1FS gasoline and gas hybrid engine, • Description of the lubrication and cooling system, • Description of the gasoline and gas fuel systems, • Description of the engine's electrical systems (ignition, starting, etc.), • Removal and refitting procedure in accordance with the manufacturer's recommendations, • Disassembly and reassembly procedure in accordance with the manufacturer's recommendations, • Procedure for checking engine components in accordance with the manufacturer's recommendations. ► **Practical part:** • Remove the 1FS Gasoline and Gas engine according to the manufacturer's recommendations, • Disassemble the engine according to the manufacturer's recommendations, • Clean the components of the Gasoline and Gas hybrid engine 1FS, • Visually inspect the components, • Check component conformity in accordance with the manufacturer's recommendations, • Replace components that do not comply with the manufacturer's recommendations, • Reassemble the components in accordance with the manufacturer's recommendations, • Refit the 1FS gasoline and gas engine in accordance with the manufacturer's recommendations, • Test and adjust the engine on the machine**.** ► **Evaluation part:** • Initial Test (Assessment of prerequisites). • Final Test (Assessment of Acquired skills).		
Target audience	Technician carrying out repairs		
Prerequisites	Have acquired the skills of a JUNIOR Technician (Preventive Maintenance)		
Language	French		
Equipment	Fully-equipped training room and workshop, TOYIE forklift truck(s) fitted with Hybrid Gasoline and Gas engine1FS, Hybrid Gasoline and Gas engine 1FS on bench.		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation.		

TOY08	REPAIR OF GASOLINE AND GAS HYBRID ENGINES (4Y-ECS WITH ELECTRONIC MANAGEMENT)

Level, mode & duration	SENIOR	Face-to-face	05 days
Trainer	**Thierry KOUGHA**		
Objectives	• **Remove** and **refit** the 4Y-ECS gasoline and gas hybrid engine in accordance with the manufacturer's recommendations, • **Disassemble** and **reassemble** the 4Y-ECS gasoline and gas hybrid engine in accordance with the manufacturer's recommendations, • **Check** component conformity in accordance with the manufacturer's recommendations, • **Test** and **adjust** the 4Y-ECS gasoline and gas hybrid engine in accordance with the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical part:** • Reminder of specific safety measures in the repair workshop, • Description and operation of the 4Y-ECS gasoline and gas hybrid engine, • Description of the lubrication and cooling systems, • Description of the gasoline and gas fuel systems, • Description of the engine's electrical system (ignition, starting, etc.), • Architecture and operation of the 4Y-ECS electronic engine management system, • Removal and refit procedure in accordance with the manufacturer's recommendations, • Disassembly and reassembly procedure in accordance with the manufacturer's recommendations, • Manufacturer's procedure for checking engine components. ► **Practical part:** • Disassemble the 4Y-ECS gasoline and gas engine according to the manufacturer's recommendations, • Disassemble the engine according to the manufacturer's recommendations, • Clean the components of the 4Y-ECS gasoline and gas hybrid engine, • Visually inspect the components, • Check that the components comply with the manufacturer's recommendations, • Replace components that do not comply with manufacturers' expectations, • Reassemble the components in accordance with the manufacturer's recommendations, • Refit the 4Y-ECS gasoline and gas engine, • Test and adjust the engine on the machine. ► **Assessment part:** • Initial Test (Assessment of prerequisites). • Final Test (Assessment of Acquired skills).		
Target audience	Technician carrying out repairs		
Prerequisites	Have acquired the skills of a JUNIOR Technician (Preventive Maintenance)		
Language	French		
Equipment	Fully-equipped training room and workshop, TOYIE forklift trucks(s) fitted with the 4Y-ECS gasoline and gas hybrid engine, Gasoline and gas hybrid engine 4Y-ECS on bench.		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation.		

TOY09	1ZS DIESEL ENGINE REPAIR

Level, mode & duration	SENIOR	Face-to-face	05 days
Trainer	**Thierry KOUGHA**		
Goals	• **Remove and refit** the 1ZS Diesel engine following the manufacturer's instructions, • **Disassemble and reassemble** the 1 ZS Diesel engine following the manufacturer's instructions, • **Check** the conformity of components to the manufacturer's expectations, • **Test and adjust** the 1ZS Diesel engine in accordance with the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical Part:** • Reminder of specific safety measures in the repair workshop, • Architecture and operation of the 1ZS “common rail” diesel engine with electronic injection, • Description of the lubrication and cooling system, • Description of fuel supply systems, • Description of the engine's electrical systems (preheating, starting, charging, etc.), • Procedure for setting the distribution and injection pump according to the manufacturer's instructions, • Procedure for removal, refitting, disassembly, reassembly following the manufacturer's instructions, • Procedure for checking engine components according to the manufacturer's instructions. ► **Practical Part:** • Remove the 1ZS Diesel engine following the manufacturer's instructions, • Disassemble the engine following the manufacturer's instructions, • Clean the components of the 1ZS Diesel engine, • Check the conformity of components to manufacturer expectations, • Replace components that do not meet manufacturer expectations, • Reassemble the components following the manufacturer's instructions, • Refit the 1ZS Diesel engine following the manufacturer's instructions, • Test and Adjust the Engine on the Machine. ► **Part Assessment:** • Initial Test (Assessment of prerequisites). • Final Test (Assessment of Acquired skills)		
Target audience	Technician performing the repair		
Prerequisites	Have acquired the skills of the JUNIOR Technician Level (Preventive Maintenance)		
Language	French		
Equipment	Equipped training room and workshop, TOYIE trolley(s) equipped with a variant of the 1ZS Diesel engine, A variant of the 1ZS engine on the bench.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		

TOY10	REPAIR OF DIESEL 1DZ ENGINE II-III,

Level, mode & duration	SENIOR	Face-to-face	05 days
Trainer	**Thierry KOUGHA**		
Objectives	• **Remove** and **refit** the diesel 1DZ engine in accordance with the manufacturer's recommendations, • **Disassemble** and **reassemble** the diesel 1DZ engine in accordance with the manufacturer's recommendations, • **Check** component conformity in accordance with the manufacturer's recommendations, • **Test** and **adjust** the diesel 1DZ engine in accordance with the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical part:** • Reminder of specific safety measures in the repair workshop, • Architecture and operation of the diesel 1DZ engine and its different variants, • Description of the lubrication and cooling systems, • Description of the fuel supply systems, • Description of the engine's electrical systems (pre-heating, starting, charging, etc.), • Procedure for timing and injection pump adjustment in accordance with the manufacturer's recommendations, • Procedure for removal, refit, disassembly and reassembly of components in accordance with the manufacturer's recommendations, • Manufacturer's procedure for checking engine components. ► **Practical part:** • Remove the diesel 1DZ engine according to the manufacturer's instructions, • Disassemble the engine according to the manufacturer's recommendations, • Clean the components of the diesel 1DZ engine, • Visually inspect the components, • Check component compliance with the manufacturer's recommendations, • Replace components that do not comply with the manufacturer's recommendations, • Reassemble the components in accordance with the manufacturer's recommendations, • Refit the diesel 1DZ engine in accordance with the manufacturer's recommendations, • Test and adjust the engine on the machine. ► **Evaluation part:** • Initial Test (Assessment of prerequisites). • Final Test (Assessment of Acquired skills)		
Target audience	Technician carrying out repairs		
Prerequisites	Have acquired the skills of a JUNIOR Technician (Preventive Maintenance)		
Language	French		
Equipment	Fully equipped training room and workshop, TOYIE forklift truck(s) fitted with a variant of the diesel1DZ engine, a variant of the 1DZ engine on a bench.		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation.		

TOY11	3Z DIESEL ENGINE REPAIR

Level, Mode and Duration	SENIOR	Face-to-face	05 days
Trainer	**Thierry KOUGHA**		
Goals	• **Removing and Reinstalling** the 3Z Diesel Engine respecting the manufacturer's recommendations, • **Disassemble and reassemble** the 3Z Diesel engine following the manufacturer's instructions, • **Check** the conformity of components to the manufacturer's expectations, • **Test and adjust** the 3Z Diesel engine in accordance with the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical Part:** • Reminder of specific safety measures in the repair workshop, • Architecture and operation of the 3Z Diesel engine and its different variants, • Description of the lubrication and cooling system, • Description of fuel supply systems, • Description of the engine's electrical systems (preheating, starting, charging, etc.), • Procedure for setting the distribution and injection pump according to the manufacturer's instructions, • Procedure for removal, refitting, disassembly and reassembly following the manufacturer's instructions, • Procedure for checking engine components according to the manufacturer's instructions. ► **Practical Part:** • Remove the 3Z Diesel engine following the manufacturer's instructions, • Disassemble the engine following the manufacturer's instructions, • Clean the components of the 3Z Diesel engine, • Visually inspect the components, • Check the conformity of components to manufacturer expectations, • Replace components that do not meet manufacturer expectations, • Reassemble the components following the manufacturer's instructions, • Refit the 3Z Diesel engine following the manufacturer's instructions, • Test and Adjust the Engine on the Machine. ► **Part Assessment:** • Initial Test (Assessment of prerequisites). • Final Test (Assessment of Acquired skills)		
Target audience	Technician performing the repair		
Prerequisites	Have acquired the skills of the JUNIOR Technician Level (Preventive Maintenance)		
Language	French		
Equipment	Equipped training room and workshop, TOYIE trolley(s) equipped with a variant of the 3Z Diesel engine, A variation of the 3Z Engine on bench.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		

TOY12	HYDRAULICS REPAIR OF ELECTRIC AND THERMAL FORKLIFT TRUCKS

Level, mode & duration	SENIOR	Face-to-face	05 days
Trainer	**Thierry KOUGHA**		
Objectives	• **Remove** and **refit** the hydraulic system organs of a TOYIE forklift truck in accordance with the manufacturer's recommendations, • **Disassemble** and **reassemble** the hydraulic system organs on a TOYIE forklift truck in accordance with the manufacturer's recommendations, • **Check** that the hydraulic components comply with the manufacturer's recommendations, • **Test and adjust** the hydraulic components and hydraulic circuit parameters.		
Description Theory (40%) Practical (60%)	► **Theoretical part:** • Basic architecture of the hydraulic systems of TOYIE Electric and Thermal forklift trucks and interpretation of hydraulic diagrams, • Description and operation of the hydraulic system of a TOYIE forklift Truck and its basic components (Pump, Motor, Distributor and Cylinders, ...). • Procedure for removal, refit, disassembly and reassembly of components in accordance with the manufacturer's recommendations, • Procedure for checking hydraulic components and the hydraulic circuit in accordance with the manufacturer's recommendations. ► **Theoretical part:** • Remove hydraulic components in accordance with the manufacturer's recommendations, • Disassemble the hydraulic components in accordance with the manufacturer's recommendations, • Clean the hydraulic components, • Check that the hydraulic components comply with the manufacturer's recommendations, • Reassemble the hydraulic components in accordance with the manufacturer's recommendations, • Refit the hydraulic components, • Test and adjust the hydraulic components in the hydraulic circuit. ► **Evaluation part:** • Initial Test (Assessment of prerequisites). • Final Test (Assessment of Acquired skills)		
Target audience	Technician carrying out repairs		
Prerequisites	Have acquired the skills of a JUNIOR Technician (Preventive Maintenance)		
Language	French		
Equipment	Equipped training room and workshop, adapted TOYIE forklift truck(s) available, SST		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation.		

TECHNICIAN EXPERT

DIAGNOSYS



TOY13	ELECTRICAL/ELECTRONIC DIAGNOSTIC OF ELECTRIC FORKLIFT TRUCKS

Level, mode & duration	EXPERT	Face-to-face	05 days
Trainer	**Thierry KOUGHA**		
Objectives	• **Diagnose** faults relating to electrical and electronic equipment on electric counterbalance trucks, • **Recommend** repair actions.		
Description Theory (40%) Practical (60%)	► **Theoretical Part :** • Technical Description of the Electric Counterbalance Truck ranges, • Electrical/Electronic Architecture of Electric Counterbalance Trucks (Interconnection of the various electronic modules on the truck's CAN network), • Electrical system malfunctions (truck ECU, ACT, ACH, BCU, etc.), • Diagnostic steps and function of the CB TRUCKCOM diagnostic tool (Component calibration, Machine parameter setting, Module programming), ► **Practical part:** 1. Collect the various symptoms of the truck malfunction/fault and the history of truck breakdowns and repairs (customer + CB TRUCKCOM tool), 2. Hypothesize the cause of the truck's malfunction/fault, 3. Test or check the conformity of the truck system and components (using the CB TRUCKCOM tool), 4. Identify the most likely hypothesis based on the tests and symptoms, 5. Elaborate a list of components to be replaced or repaired, 6. Repair the faulty component or organ, 7. Test and adjust the system to be repaired (using the CB TRUCKCOM tool). ► **Evaluation part :** • Initial Test (Assessment of prerequisites) • Final Test (Assessment of Acquired skills)		
Target audience	Technicians carrying out diagnostics		
Prerequisites	Have acquired the skills of a SENIOR Technician (Repair)		
Language	French		
Equipment	Fully-equipped training room and workshop, adapted computer(s), Adapted forklift truck(s), CASTOR diagnostic interface		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation		

TOY14	ELECTRICAL/ELECTRONIC DIAGNOSTIC OF THERMAL FORKLIFT TRUCKS

Level, mode & duration	EXPERT	Face-to-face	05 days
Trainer	**Thierry KOUGHA**		
Objectives	• **Diagnose** faults relating to electrical and electronic equipment on thermal counterbalance trucks, • **Recommend** repairs.		
Description Theory (40%) Practical (60%)	► **Theoretical Part :** • Technical Description of the Thermal Counterbalance Truck ranges, • Electrical/Electronic Architecture of Thermal Counterbalance Trucks (Interconnection of the various electronic modules on the truck's CAN network), • Description and operation of Electronic Management Systems for Thermal Counterbalance Trucks, • Electrical system malfunctions (ECU Machine and ECU ENGINE), • Diagnostic steps and function of the DENSO DST-I Diagnostic tool (Component calibration, Machine parameter setting, Module programming). ► **Practical part:** 1. Collect the various symptoms of the truck malfunction/fault and the history of truck breakdowns and repairs (customer + DENSO DST-I tool), 2. Hypothesize the cause of the malfunction/fault, 3. Test or check the conformity of the truck system and components (using the DENSO DST-I tool), 4. Identify the most likely hypothesis based on the tests and symptoms, 5. Elaborate a list of components to be replaced or repaired, 6. Repair the faulty component or organ, 7. Test and adjust the system to be repaired (using the DENSO DST-I tool). ► **Evaluation part :** • Initial Test (Assessment of prerequisites) • Final Test (Assessment of Acquired skills)		
Target audience	Technicians carrying out diagnostics		
Prerequisites	Have acquired the skills of a SENIOR Technician (Repair)		
Language	French-English		
Equipment	Fully equipped training room and workshop, adapted computer(s), Adapted forklift truck(s), DENSO DST-I diagnostic interface		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation		